

Data & AI

2. Setting Up Your Coding Agent

Dasaem Jeong · Art & Technology, Sogang University

Today

- What a coding agent is, and which model to run it on
- Every one of you leaves with a working coding agent
- Bring out your laptop now, we do this together

What is a coding agent?

Chat vs. Agent

Asking a chatbot for code

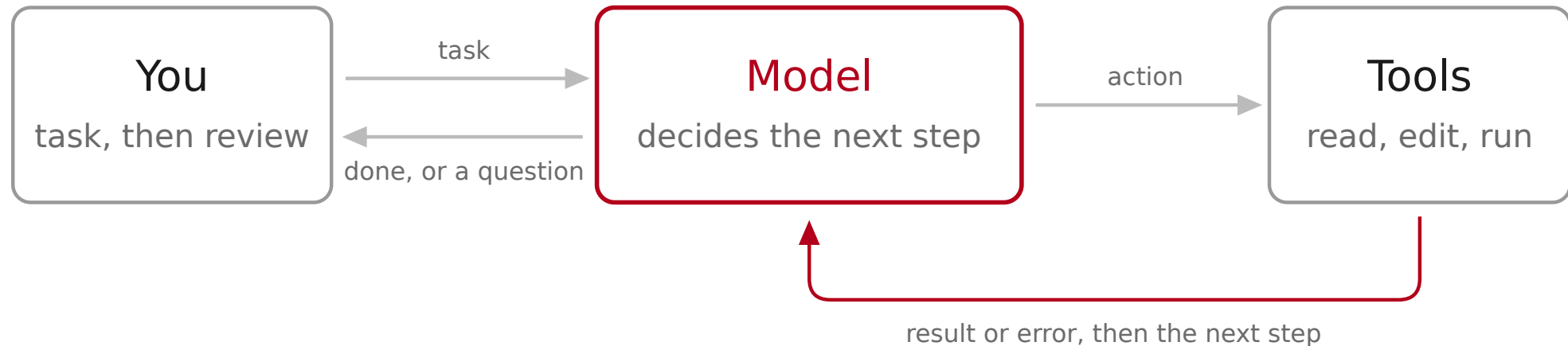
- You describe the problem and paste your code
- It answers with code
- You run it, and paste the error back

Running a coding agent

- It reads your files by itself
- It writes the code and runs it
- It reads the error and tries again

The Agent Loop

A model with tools, running until the task is done



- Tools: read a file, edit a file, run a command
- Autonomous: it picks the next action itself, and asks you only when it has to

Which model should the agent run on?

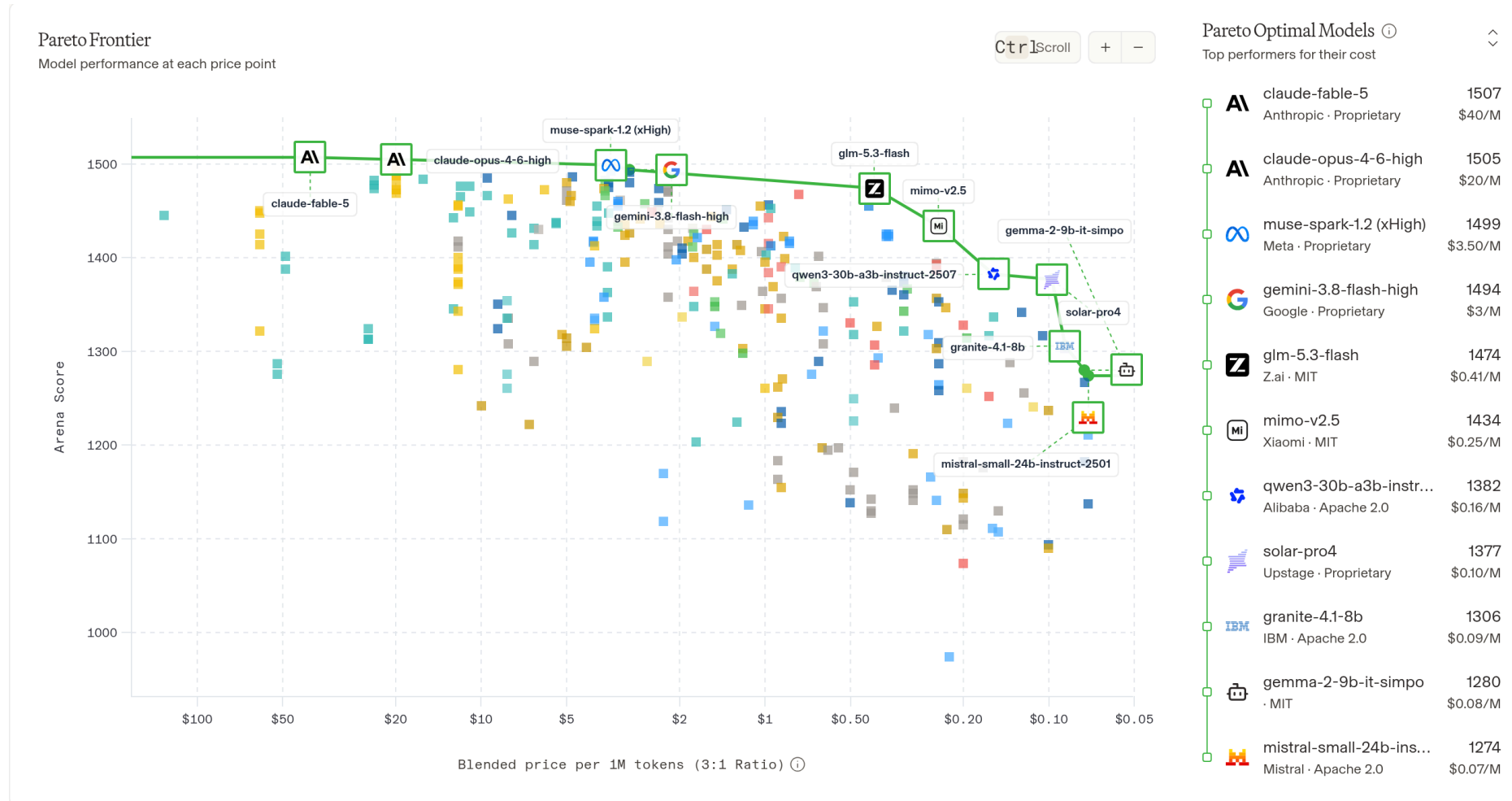
Arena

lmarena.ai

Rank 	Rank Spread 	Model 	Score 	Votes 	Price \$/M  	Context 
1	1  6	 claude-fable-5 Anthropic · Proprietary	1507 ± 5	27,189	\$10 / \$50	1M
2	1  6	 claude-opus-4-6-high Anthropic · Proprietary	1505 ± 4	72,099	\$5 / \$25	1M
3	1  14	 claude-fable-5.1-max Anthropic · Proprietary	1504 ± 11	2,906	\$10 / \$50	1M
4	1  9	 claude-opus-4-7-high Anthropic · Proprietary	1502 ± 4	60,103	\$5 / \$25	1M
5	1  17	 muse-spark-1.2 (xHigh) Meta · Proprietary	1499 ± 10	3,240	\$1.25 / \$4.25	N/A
6	3  11	 claude-opus-4-6 Anthropic · Proprietary	1498 ± 3	76,015	\$5 / \$25	1M
7	3  14	 claude-opus-4-7 Anthropic · Proprietary	1494 ± 4	61,238	\$5 / \$25	1M
8	1  22	 gemini-3.8-flash-high Google · Proprietary	1494 ± 9  Preliminary	5,125	\$0.75 / \$3.75	1M
9	4  17	 claude-opus-5-high Anthropic · Proprietary	1493 ± 5	35,174	\$5 / \$25	1M

Score Against Price

Pareto frontier on Imarena.ai



Model for This Course

deepseek-v4-flash

- 71 points below the top, and more than 20 times cheaper
- Open weights, so the same model is served by many providers
- Also cheap and good on the Gateway: gpt-5.6-luna, gemini-3.7-flash

Two Strategies

Price per million tokens, input / output, from the Arena table

Anthropic

- Aims at the top of the ranking
- Claude Fable 5.1: \$10 / \$50

OpenAI

- Aims at a good score for a low price
- GPT-5.6 Luna: \$0.20 / \$1.20

If You Pay for One Plan

20 dollars a month

Plan	Agent	What happens to the quota
ChatGPT Plus	Codex with GPT - 5.6 Luna	Luna is cheap, so the quota goes a long way
Claude Pro	Claude Code	Runs out after a few long sessions

- Claude Code makes sense on Claude Max, at 100 or 200 dollars a month
- For 20 dollars, ChatGPT Plus and Codex is the pick

Model by Task

How I split my own work

Task	Model
Chores that need a lot of reading and little thinking	GPT - 5.6 Luna
Everyday coding	Claude Opus 5
Problems where I need the best answer	Claude Fable 5.1

Three ways in, one result

Which Path Is Yours

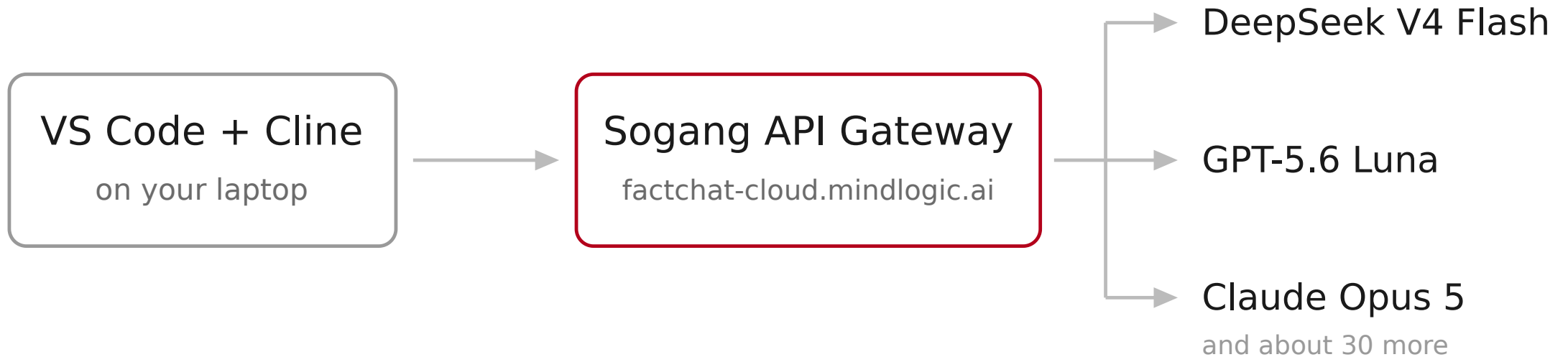
Pick the row that matches what you already have

What you already pay for	Agent	Model	What you need today
Nothing	VS Code + Cline	deepseek-v4-flash	A Sogang API Gateway key
ChatGPT Plus or Pro	Codex CLI	GPT-5.6 Luna	Sign in with your ChatGPT account
Claude Pro or Max	Claude Code	Sonnet 5 or Opus 5	Sign in with your Claude account

What is the Sogang API Gateway?

One Key, Many Models

docs.mindlogic.ai/docs/sogang/api-gateway



- Claude, GPT, Gemini, DeepSeek and more behind a single address
- Usage is charged to the account that made the key

Five Steps

- 1 Install Visual Studio Code
- 2 Create your API key
- 3 Install the Cline extension
- 4 Point Cline at the Gateway
- 5 Run one task end to end

Step 1. Install Visual Studio Code

Visual Studio Code

code.visualstudio.com

- Download the build for your operating system, then run it
- Windows: keep "Add to PATH" checked in the installer
- macOS: move the app into the Applications folder

Step 2. Create your API key

API Key

factchat-cloud.mindlogic.ai

- Sign in, then open API Gateway at the bottom of the left sidebar
- Press Create API Key and copy the key once
- The key is never shown again, so paste it somewhere safe now

Step 3. Install the Cline extension

Cline

Extensions view: Ctrl+Shift+X on Windows, Cmd+Shift+X on macOS

- Search for Cline and press Install
- A Cline icon appears in the left activity bar
- Open it, and choose the option to use your own API key

Step 4. Point Cline at the Gateway

Cline Settings

Gear icon, then API Provider

Field	Value
API Provider	OpenAI Compatible
Base URL	<code>https://factchat-cloud.mindlogic.ai/v1/gateway</code>
API Key	the key you copied in step 2
Model ID	<code>deepseek-v4-flash</code>

- Pick OpenAI Compatible, not OpenAI

Model Choice

Model ID is a plain text field

- `deepseek-v4-flash` is the default for this course
- `gpt-5.6-luna` and `gemini-3.7-flash` also work
- Change one line here to change the model

Step 5. Run one task end to end

First Task

Open a folder first, then type this into Cline

```
Create hello.py that prints today's date,  
then run it and show me the output.
```

- Cline asks before it writes a file or runs a command
- If the file appears and the date is right, you are set up

Plan and Act

- Plan mode writes a plan, Act mode changes your files
- Read the diff before you approve it

If you already have Claude Code

Claude Code

~/.claude/settings.json

```
{  
  "env": {  
    "ANTHROPIC_BASE_URL": "https://factchat-cloud.mindlogic.ai/v1/gateway/claude",  
    "ANTHROPIC_AUTH_TOKEN": "YOUR_API_KEY"  
  }  
}
```

- Use ANTHROPIC_AUTH_TOKEN, not ANTHROPIC_API_KEY
- Reload the VS Code window after you edit the file

If you already have Codex CLI

Codex CLI

~/.codex/config.toml

```
model = "gpt-5.6-luna"  
model_provider = "factchat"
```

[model_providers.factchat]

```
name = "FactChat"  
base_url = "https://factchat-cloud.mindlogic.ai/v1/gateway"  
env_key = "FACTCHAT_API_KEY"  
wire_api = "responses"
```

- Put the key in FACTCHAT_API_KEY in your shell

When it does not work

Reading the Error

Message	Cause	Fix
401	Wrong key, or a stray space	Paste the key again
404	Base URL has an extra path	Stop the URL at /v1/gateway
429	Too many requests in a minute	Wait, then retry
Model not found	Typo in the Model ID	Use deepseek-v4-flash
401 with api.openai.com	Provider is OpenAI	Switch it to OpenAI Compatible

Your Key Is Yours

- Never put the key in a file you push to GitHub
- If it leaks, revoke it on the API Gateway page and make a new one

Before You Leave

- Visual Studio Code is installed
- Cline answers you through the Gateway
- One task ran from prompt to output

Q & A